Adding Homes Gradually Across the Suburb

Strategies for modelling distributed housing growth over time in the Walkerville sewer model

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1. Purpose

At the 28 September meeting and the 26 October review the industry partner made two points about growth: homes do not appear in one day but take years from approval to connection, and infill is usually spread across the network rather than placed in one spot [1, 2]. This report sets out the options for doing that in our model, shows with figures what each option changes, and recommends a way to run them. All maps use the real council network (887 pipes, 328 manholes). Flows in the figures are worked out by adding up the new homes upstream of each manhole; no new SWMM runs were made for this report.

2. What the model does now

- **Sim 3** grows one of eight areas at a time (R1 to R8, Figure 1) on a build-out ladder from 10% to 300% of that area's share of the PlanSA 2041 high case. New homes are shared over the area's existing connections, in proportion to the homes already on each pipe [8].
- **Sim 3-e** grows the whole suburb at once with the same sharing rule and turns build-out into years by assuming the homes arrive evenly up to 2041 [8].

Within the modelled area the 2041 projection is about **436 homes** in the high case and **351** in the medium case (the council totals of 485 and 390, scaled by the 2,565 of 2,851 connections the model holds) [3, 8]. At the peak hour each new home adds 0.0106 L/s (2.3 people, 200 L a day, peak factor 2.0, as set in the team's settings file [8]), so all 436 together add about **4.6 L/s** at the outlet, against roughly 100 L/s already flowing there in the Design wet case.

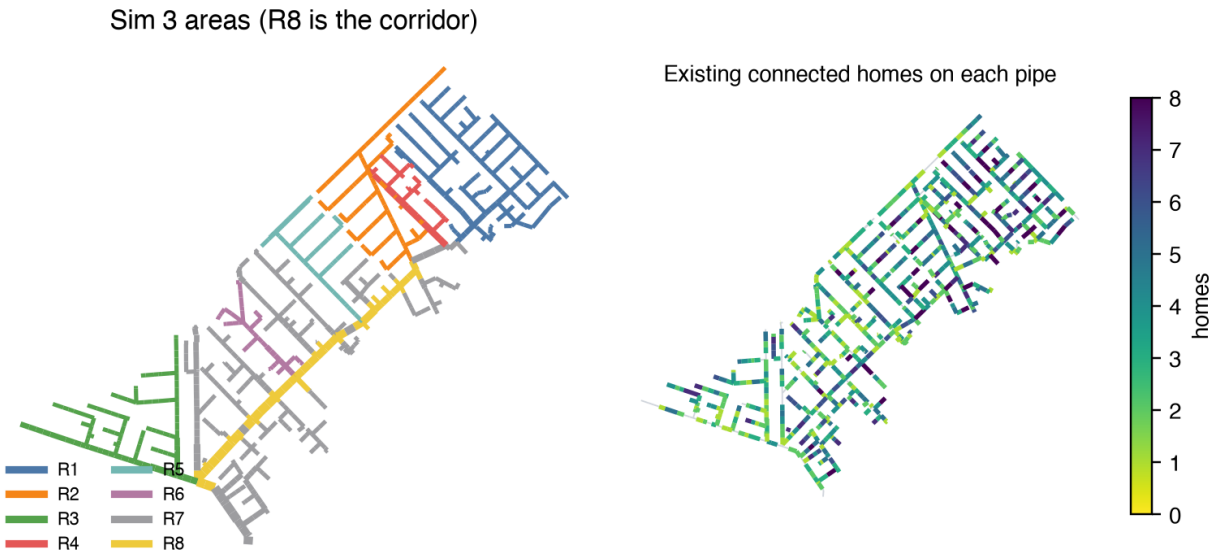


Figure 1: Left: the eight Sim 3 areas. Right: homes already connected to each pipe, which the current rule uses to share out new homes.

3. Two separate choices: where and when

Distributed growth over time is really two decisions that can be made separately:

- **Where** the new homes connect (spatial strategy, S1 to S5).
- **When** they arrive between now and 2041 (time profile, T1 to T4).

Any spatial strategy can be paired with any time profile. This matters for the cost of running them, as Section 8 explains.

4. Where: spatial strategies

Strategy	How the homes are placed	Strengths and weaknesses
S1 Proportional (current rule)	Every pipe gets a share equal to its share of existing homes.	Simple and repeatable, and already run in Sim 3 and Sim 3-e [8]. Matches the partner's advice that infill is usually spread [2]. Spreads homes more evenly than real infill does, so no branch ever gets a large jump.
S2 Corridor-weighted	A set share (50% in the figure) goes to pipes along Walkerville Terrace; the rest follows S1.	The projections name a Walkerville Terrace code amendment as a growth driver [3]. The 50% is my choice, not a figure from PlanSA, so it should be shown as a range.
S3 Random scatter	Each new home is given to one existing connection at random, with the chance in proportion to S1. Repeated many times.	Shows the range of outcomes rather than one answer, in the same way that Monte Carlo sampling is used for other unknown inputs [10]. On average it equals S1. Earlier studies also ran many development scenarios rather than one [12, 14]. Needs many runs.
S4 Clustered sites	A few sites (5 in the figure) take half the homes, for example a block of units; the rest follows S1.	Closest to how a large development lands, and similar to the concentrated scenarios in [11]; the partner noted that greenfield growth is geographically concentrated [2], and Torbay checked local sewers development by development [7]. Site locations are unknown, so they are drawn at random and repeated.

Strategy	How the homes are placed	Strengths and weaknesses
S5 From real data	Homes placed by zoning, lot size and development approvals.	The most realistic, and closest to parcel-based growth models [13]. PlanSA reports that most demolitions for infill are of 1950s and 1960s houses on sites with a low capital to site value ratio [5], and lists Walkerville among suburbs with notable net growth [4]. Needs the council's zoning layer and approval records, which we do not have yet.

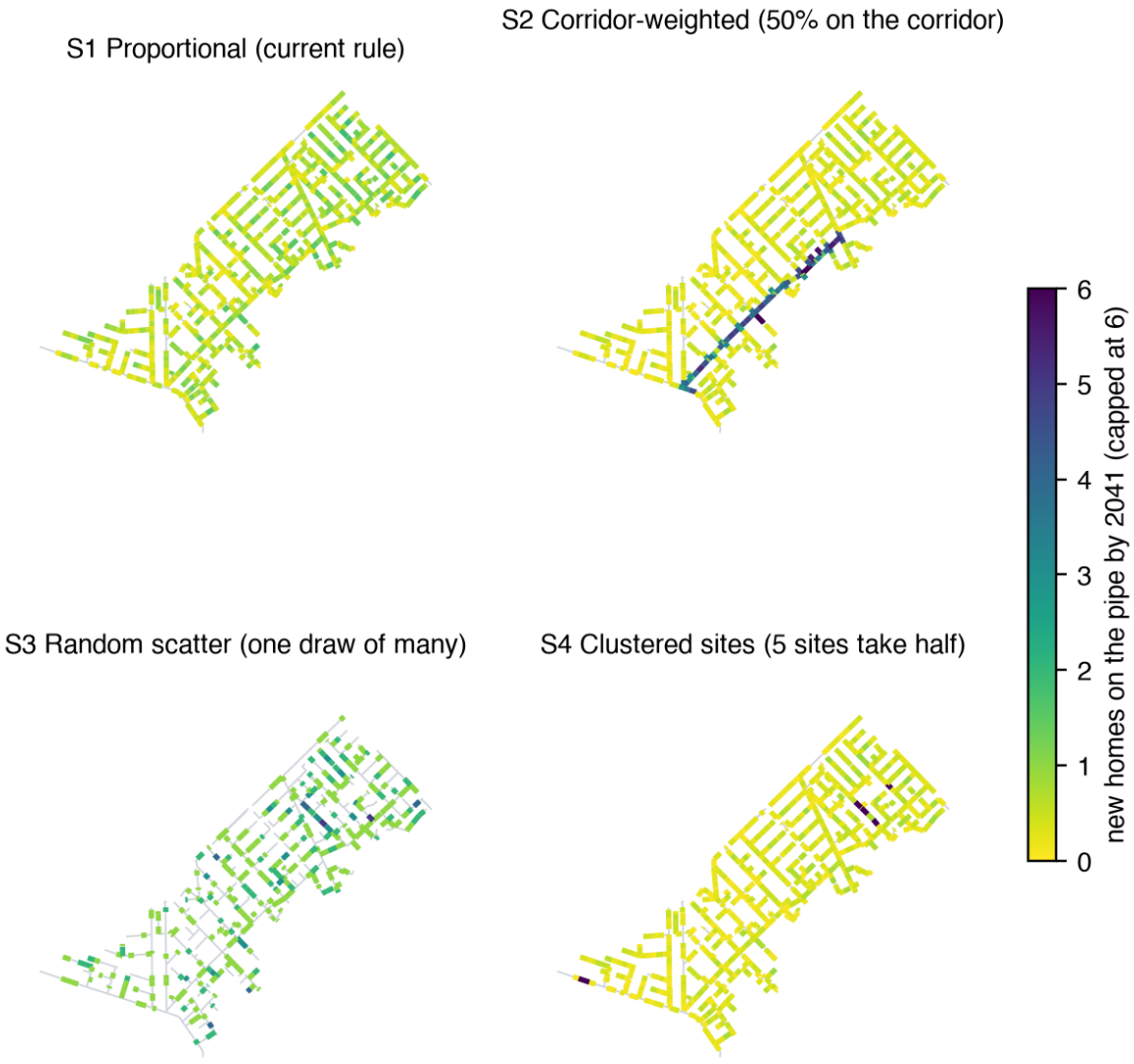


Figure 2: Where the 436 high-case homes land by 2041 under S1 to S4. Grey pipes receive none. S3 and S4 show one random draw each; every draw is different.

5. When: time profiles

Profile	Description
T1 Even (current rule)	The same number of homes each year. The simplest assumption; PlanSA gives one 2041 total [3], and Sim 3-e uses even arrival [8].
T2 S-curve	Slow at first, fastest in the early 2030s, then slowing. An illustrative shape, not taken from a source.
T3 Early surge	Fast at first, then slowing. An illustrative "growth comes sooner than expected" case, not taken from a source.
T4 Batches	Homes arrive in lumps (every 4 years in the figure), as approvals and larger projects are completed. The partner said a home takes about three to five years from approval to connection [1], or five to six years to build [2], and the state sets council housing targets in 10-year steps [6].

The shapes of T2 to T4 are illustrations chosen for this report, not projections. All four reach the same total in 2041, and each can be run for both the high and medium cases. Testing more than one path follows the use of several transition paths in [14].

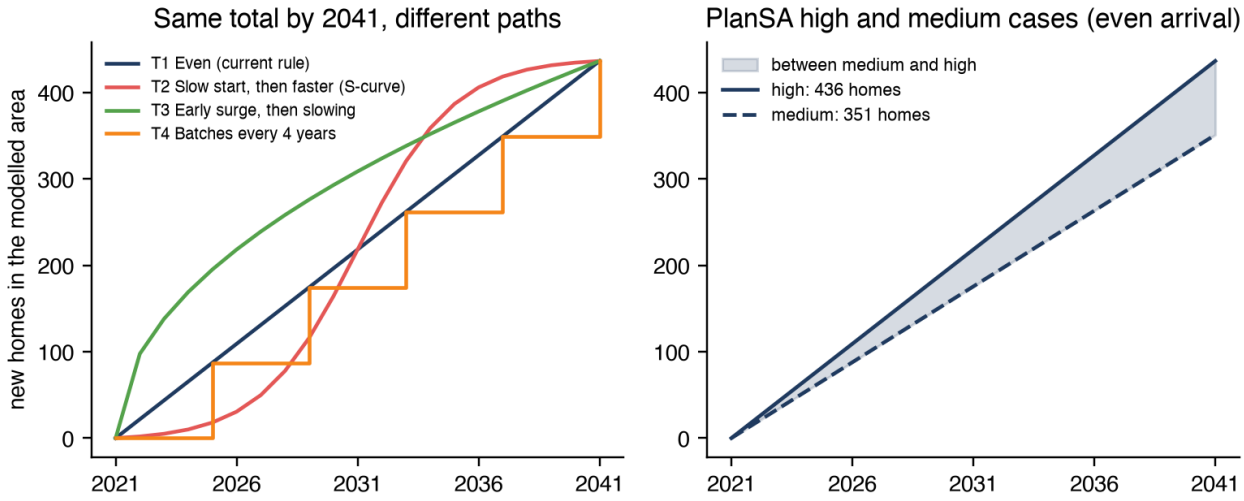


Figure 3: Left: four ways of reaching the same 2041 total. Right: the gap between the PlanSA high and medium cases under even arrival.

6. What earlier studies did

No published study was found that spreads housing growth over a separate sewer network in exactly these four ways. The strategies do, however, follow approaches that earlier urban drainage studies have used.

- **Spread against concentrated growth.** Mikovits et al. [11] linked an urban development model to a calibrated SWMM model of Innsbruck and compared the same population growth placed three ways: spread over many small sites across the city, or concentrated in one of two large areas. The spread scenario gave clearly larger increases in flooding and overflow volume. This is the closest precedent for comparing S1 and S3 (spread) with S2 and S4 (concentrated).
- **Many runs instead of one forecast.** The same group designed their urban development model to run "a manifold of simulations" with little input data, so that the possible range of changes is covered rather than one prediction [12]. Van der Werf et al. [14] modelled gradual changes in two Dutch catchments using both "probabilistic and projected transitional paths". These support repeating random draws (S3 and S4) and running more than one time path (T1 to T4).

- **Location matters, not only totals.** Mikovits et al. [13] placed buildings and population parcel by parcel in a "dynamic, cyclic process of automated city growth" and found that population totals alone were not enough to describe the effect on drainage. Parcel-based placement of this kind is the model for S5.
- **Densifying existing areas.** Mahaut and Andrieu [16] compared densifying existing urban areas with expanding into new ones in Nantes, and Duque et al. [15] explored many future development alternatives for a sanitary sewer network. Both treat infill of already-served areas as a distinct scenario, which is what S1 to S3 represent.

What differs in our case. The Innsbruck, Nantes and Dutch studies are mostly about combined sewers, where new development mainly adds rain runoff from new roofs and paving. Walkerville has a separate sewer, where new homes mainly add sewage, so their results on flood volumes do not carry across directly. The way they set up scenarios does.

7. What the figures show

1. On the trunk, the spatial strategy hardly matters

Flow adds up as it moves downstream, so a manhole near the outlet receives almost every new home whichever way they are spread (Figure 4). For manholes with more than 1,000 homes upstream, the typical spread in added flow between random draws is about 3% for S3 and 12% for S4. Near the outlet every strategy gives the same 4.6 L/s.

2. On small branches, it matters a great deal

For manholes with fewer than 50 homes upstream, the typical spread is about 68% for S3 and over 300% for S4. At the branch manhole MH1625959 (40 homes upstream), 90% of S3 draws give between 0.44 and 1.62 times the S1 value. Under S4, most draws give half the S1 value, because no cluster lands there, but in a few draws a cluster does land and the added flow is up to about 7 times larger. S2 halves the added flow on every branch away from the corridor and raises it by up to about 9 times on corridor pipes.

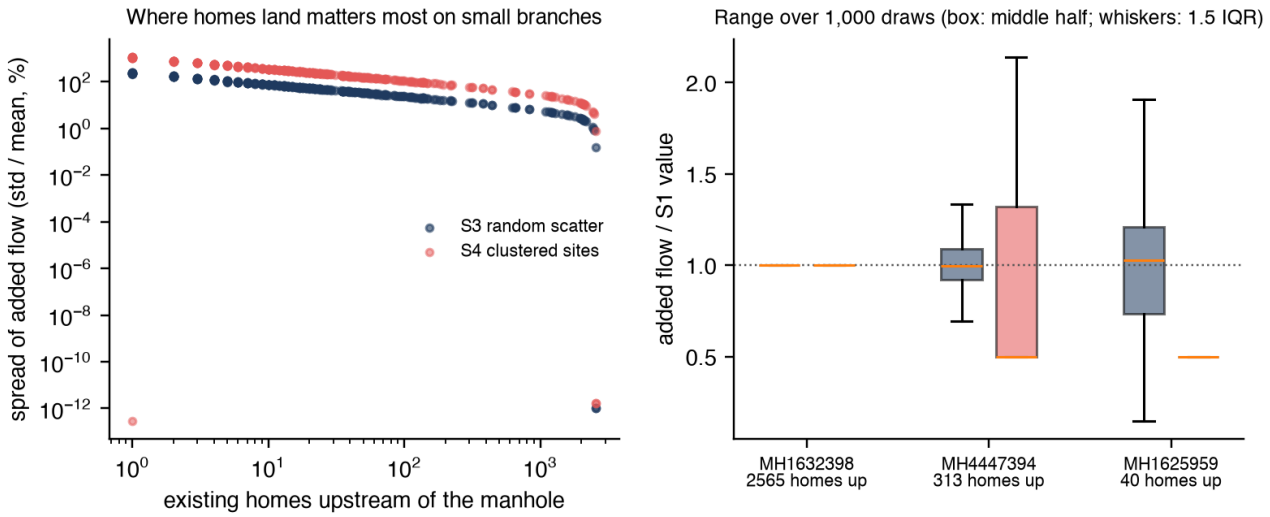


Figure 4: Left: spread of added flow over 1,000 random draws, against the number of homes upstream of each manhole (both axes log scale). Right: added flow relative to S1 at three manholes, for S3 (blue) and S4 (red).

3. Timing moves the date, and spread moves the size

Figure 5 follows the branch manhole through time. The time profiles change *when* a given amount of flow arrives: under T3 the 2041 flow is reached years earlier than under T2. The spatial strategy changes *how much* arrives at all, and for a branch this effect is larger than the timing effect.

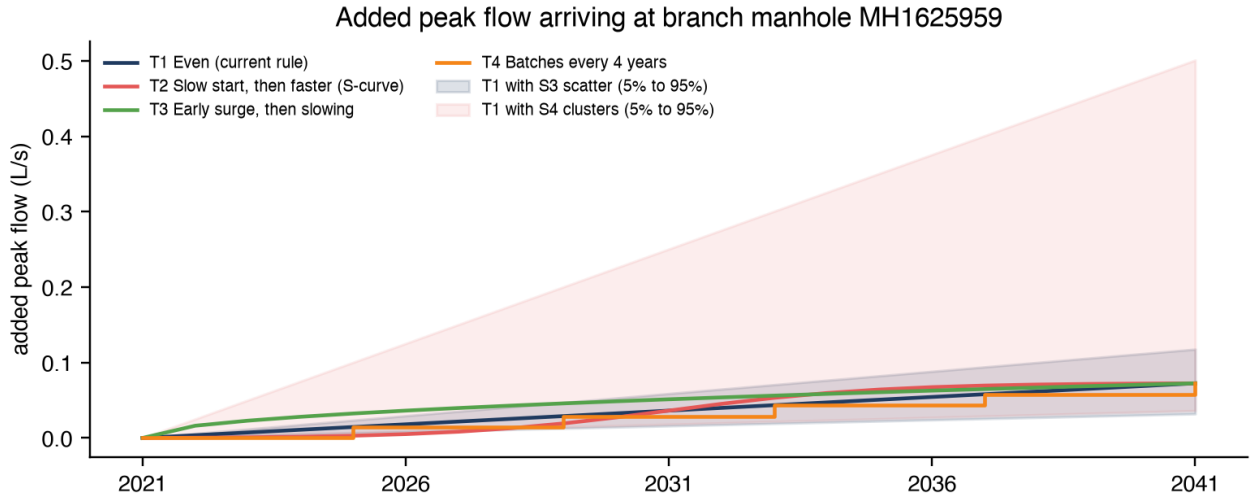


Figure 5: Added peak flow at branch manhole MH1625959 under the four time profiles with S1, and the 5% to 95% range of S3 and S4 under even arrival.

What this means for sensors. A sensor on the trunk sees the total growth reliably, whatever the pattern. A sensor on a branch may see much more or much less than S1 predicts. So a placement chosen under S1 alone may look better on branches than it really is, and testing placements against S3 and S4 draws is a fair way to check them.

8. How to run it

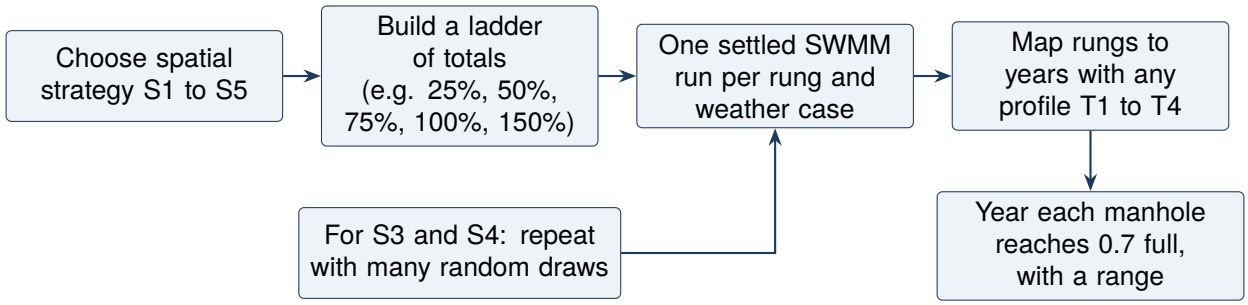


Figure 6: Suggested workflow. The time profile is applied after the runs, so it costs no extra SWMM runs.

Key point: time profiles are free. The model is run until levels settle, so it has no memory of how the homes got there. If the spatial pattern only grows in size over time (S1 and S2), the depths for a given total are the same whatever year that total is reached. One set of runs on a ladder of totals can therefore be relabelled with any time profile and either PlanSA case. Sim 3-e already works this way for even arrival [8]. For S3 and S4 the pattern itself changes as new sites appear, so each draw needs its own sequence, built so that later years keep the earlier homes.

Run budget. Sim 3-e's 56 whole-suburb runs took about 8 minutes on 8 workers, roughly one worker-minute per run [8]. On that basis:

Option	Runs	Time on 8 workers
S1 (done in Sim 3-e)	7 rungs × 8 cases = 56	about 8 min
S2 at three corridor shares	3 × 5 rungs × 8 cases = 120	about 15 min
S3, 50 draws	50 × 4 years × 2 cases = 400	about 50 min
S4, 50 draws	50 × 4 years × 2 cases = 400	about 50 min

The S3 and S4 rows use only two weather cases (Design wet and Beyond design) to keep the time reasonable; more can be added later.

A possible shortcut, not yet checked. Sim 2.5 already holds runs where homes were added at one manhole at a time, across the whole area [8]. If the rise in level from several small additions is close to the sum of the rises from each one alone, any S3 or S4 draw could be estimated from those saved runs without new SWMM runs. This is unlikely to hold where pipes run nearly full, so it would first need to be tested against about ten direct runs before it is relied on.

9. Recommendation

1. Keep **S1** as the base case, and report it with **all four time profiles** and both PlanSA cases, since this needs no new runs.
2. Add **S3** (50 draws) to give every result a range, and use it to test whether the chosen sensor sites still work when growth does not follow the average.
3. Add **S2** at a few corridor shares and **S4** as a stress case for branches.
4. Move to **S5** once the council's zoning layer and approval records are available, and use them to replace the illustrative shares and site counts.

10. Limitations of this report

- The figures add up new homes upstream of each manhole. This shows how much extra flow arrives, not how much the level rises, which depends on each pipe's size, slope and how full it already is. That needs the SWMM runs above.
- Relief pipes are treated as passing all of their flow on, and the two pump stations as passing on everything they receive, as in Sim 3.
- The 50% corridor share, the five clustered sites and the shapes of T2 to T4 are choices made for illustration. They are not taken from PlanSA or the council.
- New homes add sewage only, as in Sim 3; any extra leakage from new connections is left out.

11. Where each idea comes from

Item	Basis	Source
Growth spread over the suburb	Partner: infill "will be spread" across the network	[2]
Growth arriving over years	Partner: approval to connection takes years; "all dwellings will not appear in one day"	[1, 2]

Item	Basis	Source
436 and 351 homes	PlanSA high (485) and medium (390) cases for Walkerville, scaled to the modelled connections	[3, 8]
Flow per new home	2.3 people (ABS 2021 [9]), 200 L a day, peak factor 2.0, from the team's settings file	[8]
S1 Proportional	Sim 3's sharing rule; spread (distributed) scenario and densification of existing areas in earlier studies	[8, 11, 15, 16]
S2 Corridor-weighted	Walkerville Terrace named in the projections; corridor pipes as defined in Sim 2.5 and Sim 3; growth concentrated in a named area as in [11]. The 50% share is my choice	[3, 8, 11]
S3 Random scatter	Many simulations to cover the range of possible development; probabilistic transition paths; Monte Carlo sampling. The exact rule (one home to one random connection) is my choice	[10, 12, 14]
S4 Clustered sites	Greenfield growth is concentrated; concentrated scenarios in [11]; development-by-development checks. Five sites and 50% are my choices	[2, 7, 11]
S5 From real data	Parcel-based placement of buildings and population; infill demolitions by house age and site value; Walkerville growth	[4, 5, 13]
T1 Even	One 2041 total; Sim 3-e's assumption; projected transition paths	[3, 8, 14]
T2 S-curve, T3 Early surge	Running several time paths follows [14]; these two shapes are illustrative only	[14] (idea only)
T4 Batches	Approval-to-connection lag; targets set in 10-year steps. The 4-year interval is my choice	[1, 2, 6]
Time profiles need no new runs	Follows from the settled (steady) runs used in Sim 3 and Sim 3-e	[8]
Run times	Measured on Sim 3-e	[8]
Superposition shortcut	An idea to be tested, using the Sim 2.5 single-site runs	[8]

Sources

[1] SA Water and capstone team PG-S2-22 (2026). Industry partner meeting, 28 September 2026, transcript (V. Maruvada at 19:53, 21:48 and 34:14).

[2] SA Water and capstone team PG-S2-22 (2026). Industry partner prototype demonstration, 26 October 2026, transcript (V. Maruvada at 50:35 and 55:12).

[3] PlanSA (n.d.). *Local Area Population Projections 2021 to 2041*, Walkerville (high and medium cases; Walkerville Terrace code amendment note), as recorded in the team's assumptions register (items G14 and regions anchor).

[4] Government of South Australia (2023). *Land Supply Report for Greater Adelaide: Residential*. PlanSA.

[5] Government of South Australia (2021). *Land Supply Report for Greater Adelaide, Part 2: Urban Infill*. PlanSA.

[6] Government of South Australia (2024). *Greater Adelaide Regional Plan Fact Sheet: Housing and Land Supply Targets*. PlanSA.

[7] AECOM (2014). *Torbay 2032: An Assessment of Future Sewer Capacity in Torbay*. Report for Torbay Council and South West Water.

[8] Capstone team PG-S2-22 (2026). *gravity-sewer-osp* repository: `config.js` (loads and regions), `sim3_regions.py` (Sim 3), `sim25_grid.py` (Sim 2.5 and corridor), `sim3e/` (Sim 3-e, A. J. Kalathil). `https://github.com/AbrahamPaulJ/gravity-sewer-osp`.

[9] Australian Bureau of Statistics (2021). Census of Population and Housing, as used for household size in [8].

[10] Zheng, Y., Jin, X., Wei, J., Zhou, Y. and Zhang, Y. (2025). A novel framework for optimization and evaluation of sensors network in urban drainage system. *Water Research* 270, 122833.

[11] Mikovits, C., Rauch, W. and Kleidorfer, M. (2018). Importance of scenario analysis in urban development for urban water infrastructure planning and management. *Computers, Environment and Urban Systems* 68, 9–16.

[12] Mikovits, C., Rauch, W. and Kleidorfer, M. (2015). A dynamic urban development model designed for purposes in the field of urban water management. *Journal of Hydroinformatics* 17(3), 390–403.

[13] Mikovits, C., Rauch, W. and Kleidorfer, M. (2014). Dynamics in urban development, population growth and their influences on urban water infrastructure. *Procedia Engineering* 70, 1147–1156.

[14] van der Werf, J. A., Kapelan, Z. and Langeveld, J. (2023). The impact of blue-green infrastructure and urban area densification on the performance of real-time control of sewer networks. *Water Resources Research* 59, e2022WR033591.

[15] Duque, N., Scholten, L. and Maurer, M. (2024). Exploring transitions of sewer wastewater infrastructure towards decentralisation using the modular model TURN-Sewers. *Water Research* 257, 121640.

[16] Mahaut, V. and Andrieu, H. (2019). Relative influence of urban-development strategies and water management on mixed (separated and combined) sewer overflows in the context of climate change and population growth: A case study in Nantes. *Sustainable Cities and Society* 44, 171–182.